

Question :

Java programming -Java Buzzwords, Data types, Variables, Operators, Control structures, Arrays, Console input and output

Sample Program:

2. Write a java program which reads marks of five subjects through command line and print the total and average.

1. Steps of the Program

1. Import the Scanner class.
2. Create the Average class.
3. Create a Scanner object to take input.
4. Initialize total = 0.
5. Ask the user to enter 5 marks.
6. Use a for loop to read 5 marks.
7. Add each mark to total.
8. Calculate the average using total / 5.0.
9. Display the total marks.
10. Display the average marks.
11. Close the Scanner.
12. End the program.

2. Overall Algorithm

1. **Start**
2. Initialize total = 0.
3. Set i = 1.
4. Repeat while i <= 5:
 - Read marks.
 - Add marks to total.
 - Increment i.
5. Calculate:

Average = Total / 5.0

6. Display total.
7. Display average.
8. Close the Scanner.
9. **Stop.**

Output :

```
|_____|
|      Average      |
|_____|
| - sc : Scanner    |
| - total : int      |
| - i : int          |
| - marks : int      |
| - average : double |
|_____|
| + main(args : String[]) : void |
|_____|
```

Code :

```
import java.util.Scanner;
```

```
public class Average {
```

```
    public static void main(String[] args) {
```

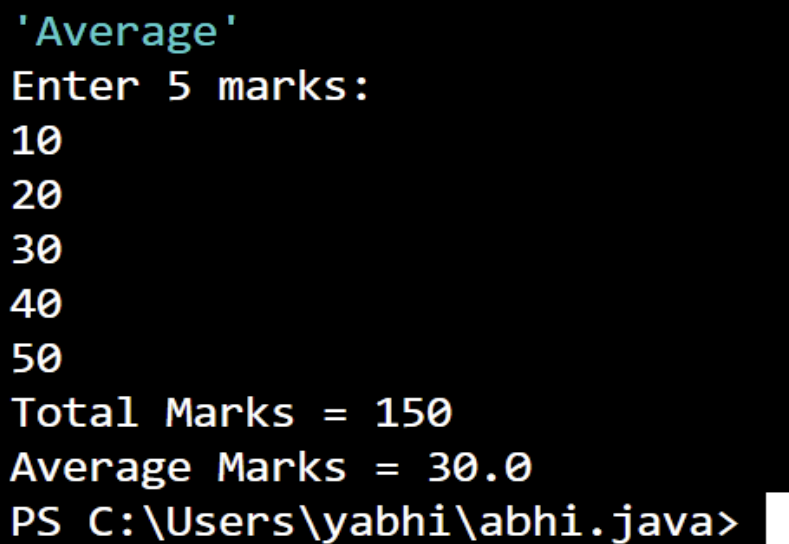
```
        Scanner sc = new Scanner(System.in);
```

```
        int total = 0;
```

```
        System.out.println("Enter 5 marks:");
```

```
for (int i = 1; i <= 5; i++) {  
    int marks = sc.nextInt();  
    total = total + marks;  
}  
  
double average = total / 5.0;  
  
System.out.println("Total Marks = " + total);  
System.out.println("Average Marks = " + average);  
  
sc.close();  
}  
}
```

Output :



The screenshot shows a terminal window with a black background and white text. The output of the Java program is displayed, including the prompt 'Average', the instruction 'Enter 5 marks:', and the five input values 10, 20, 30, 40, and 50. The final results are 'Total Marks = 150' and 'Average Marks = 30.0'. The prompt 'PS C:\Users\yabhi\abhi.java>' is visible at the bottom, followed by a white cursor block.

```
'Average'  
Enter 5 marks:  
10  
20  
30  
40  
50  
Total Marks = 150  
Average Marks = 30.0  
PS C:\Users\yabhi\abhi.java>
```